



**ALLIANCE
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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 7

Title: Developing finance KPI dashboards

Submitted By:

Name: Lavanya E

Reg. No.: 2511022250083

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Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application

Alliance University

Chandapura - Anekal Main Road, Anekal

Bengaluru – 562106

Developing Finance KPI Dashboard

Problem Statement

A manufacturing company wants to monitor its overall financial performance using an interactive Power BI dashboard. Senior management requires a single dashboard to analyze budget allocation, actual expenditure, budget variance, budget utilization, department performance, category-wise spending, regional performance, and payment methods. The objective is to design an interactive Finance KPI Dashboard using the Budget vs Actual Financial Dataset that supports effective financial monitoring and business decision-making.

Dataset Used: Budget vs Actual Financial Dataset (Kaggle)

Methodology / Steps Followed

1. **Downloaded** the Budget vs Actual Financial Dataset from Kaggle.
2. **Imported** the dataset into Power BI Desktop using the **Get Data** option.
3. **Cleaned and verified** the data using Power Query by checking data types, duplicate records, and missing values.
4. **Converted and verified** the Date, Budget Amount, and Actual Amount columns into the appropriate data types.
5. **Created financial KPIs** to analyze Total Budget, Total Actual, Budget Variance, Budget Utilization, and Total Transactions.
6. **Created KPI Card Visuals** to display Total Budget, Total Actual, Budget Variance, Budget Utilization %, and Total Transactions.
7. **Created charts** to compare Budget and Actual Amount by Department, Category, Region, and Year.
8. **Created additional visualizations** such as Payment Method analysis, Budget Variance by Department, and Budget Amount by Category.
9. **Added interactive slicers** for Year, Department, Region, Category, and Payment Method to enable customized analysis.
10. **Applied professional formatting** using a dark corporate theme, contrasting colors, consistent fonts, and proper alignment to improve dashboard readability and presentation.

Output

- Loading the Budget vs Actual Financial Dataset

Get Data (Power Query)

Choose data

Search

Display options

Excel workbook

Sheet1

Date	Department	Category	Region	Budget Amount	Actual Amount	Payment Method	Transaction ID
6/1/2022 12:00:00 AM	Sales	Travel	North	120006	43048	Cash	TEN100000
11/1/2023 12:00:00 AM	Marketing	Saleses	East	19702	47896	Bank Transfer	TEN100001
5/0/2021 12:00:00 AM	IT	Training	Central	00523	003632	Cash	TEN100002
4/12/2022 12:00:00 AM	Marketing	Saleses	North	114711	80374	Cash	TEN100003
11/27/2021 12:00:00 AM	Sales	Utilities	Central	121895	64314	UPI	TEN100004
3/29/2021 12:00:00 AM	Finance	Marketing	East	37762	70644	Bank Transfer	TEN100005
3/02/2023 12:00:00 AM	IT	Marketing	West	144211	45127	UPI	TEN100006
5/11/2021 12:00:00 AM	HR	Marketing	Central	22567	52015	Cash	TEN100007
2/9/2023 12:00:00 AM	IT	Infrastructure	West	85737	58734	UPI	TEN100008
12/10/2021 12:00:00 AM	IT	Saleses	East	16857	753432	Cash	TEN100009
3/17/2023 12:00:00 AM	IT	Utilities	East	75001	70056	Bank Transfer	TEN100010
1/01/2022 12:00:00 AM	Finance	Utilities	Central	85208	209122	Cash	TEN100011
8/14/2023 12:00:00 AM	Sales	Marketing	Central	68006	70502	Bank Transfer	TEN100012
10/4/2021 12:00:00 AM	HR	Training	East	89145	57102	UPI	TEN100013
4/5/2022 12:00:00 AM	Sales	Infrastructure	Central	42270	148677	Cash	TEN100014
1/02/2021 12:00:00 AM	Marketing	Infrastructure	Central	130071	79899	Cash	TEN100015
8/10/2021 12:00:00 AM	Marketing	Marketing	Central	12012	89817	Cash	TEN100016
1/18/2023 12:00:00 AM	Marketing	Marketing	Central	69345	120109	Cash	TEN100017
5/7/2023 12:00:00 AM	Sales	Saleses	North	54889	20752	Bank Transfer	TEN100018
4/00/2022 12:00:00 AM	Operations	Training	East	39506	30720	Cash	TEN100019
12/19/2023 12:00:00 AM	Finance	Infrastructure	East	49080	167039	Cash	TEN100020
5/26/2022 12:00:00 AM	Operations	Utilities	South	66878	743486	Bank Transfer	TEN100021
12/1/2022 12:00:00 AM	Marketing	Marketing	West	75588	71379	Cash	TEN100022

Back Cancel Load

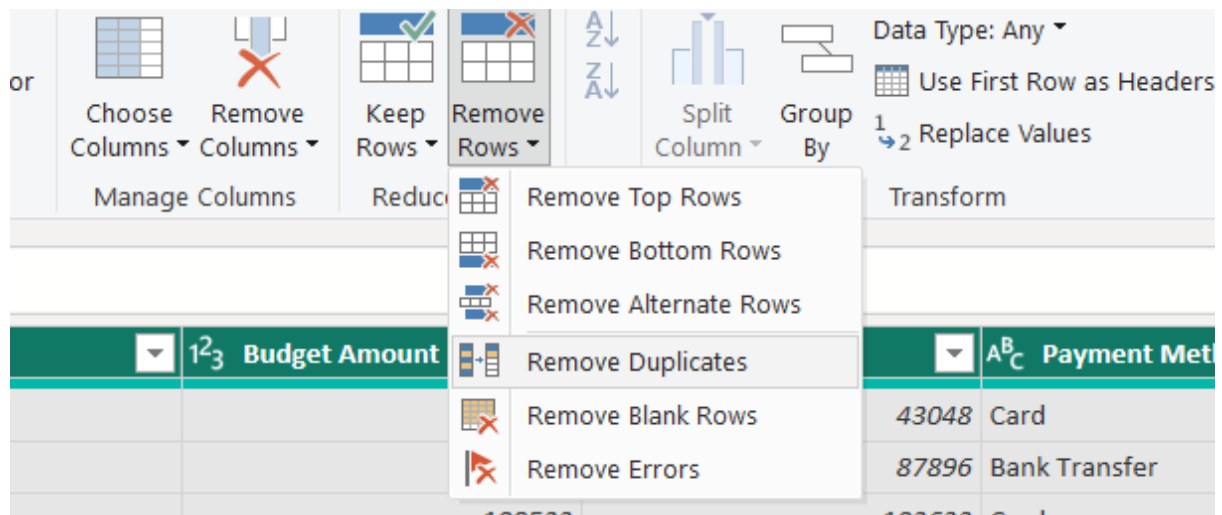
➤ Data Cleaning and Data Type Verification

Table

Any Column

Table.TransformColumns

	Date	
1	1.2	Decimal Number
2	\$	Fixed decimal number
3	123	Whole Number
4	%	Percentage
5		Date/Time
6		Date
7		Time
8		Date/Time/Timezone
9		Duration
10	ABC	Text
11		True/False
12		Binary
13		Using Locale...
15	05-04-2022 00:00:00	Sales
16	22-01-2021 00:00:00	Marketi
17	10-09-2021 00:00:00	Marketi
18	18-01-2023 00:00:00	Marketi
19	07-05-2023 00:00:00	Sales
20	20-04-2022 00:00:00	Operati
21	19-12-2023 00:00:00	Finance



► Creating Financial KPI Measures using DAX

```

1 Date Table =
2 ADDCOLUMNS(
3     CALENDAR(
4         MIN('Financial Data'[Date]),
5         MAX('Financial Data'[Date])
6     ),
7     "Year", YEAR([Date]),
8     "Month Number", MONTH([Date]),
9     "Month", FORMAT([Date], "MMM"),
10    "Year Month", FORMAT([Date], "YYYY-MM")
11 )

```

```

1 Total Budget =
2 SUM('Financial Data'[Budget Amount])

```

```

1 Total Actual =
2 SUM('Financial Data'[Actual Amount])

```

```

1 Budget Variance =
2 [Total Budget] - [Total Actual]

```

```
1 Budget Utilization % =  
2 DIVIDE(  
3     [Total Actual],  
4     [Total Budget],  
5     0  
6 )
```

```
1 Total Transactions =  
2 DISTINCTCOUNT('Financial Data'[Transaction ID])
```

```
1 Variance % =  
2 DIVIDE(  
3     [Budget Variance],  
4     [Total Budget],  
5     0  
6 )
```

```
1 Budget Status =  
2 IF(  
3     [Budget Variance] >= 0,  
4     "Under Budget",  
5     "Over Budget"  
6 )
```

```
1 Overspending Amount =  
2 IF(  
3     [Budget Variance] < 0,  
4     ABS([Budget Variance]),  
5     0  
6 )
```



Results

- The Budget vs Actual Financial Dataset was successfully imported, cleaned, and analyzed using Power BI.
- Data quality was verified by checking data types, duplicate records, missing values, and transaction information.

- KPI Cards provided an overview of total budget, total actual expenditure, budget variance, budget utilization, and total transactions.
- The Department-wise Budget vs Actual chart helped identify differences between planned and actual expenditure across departments.
- The Budget Variance chart helped identify departments with higher or lower financial variance.
- The Category-wise chart helped analyze the distribution of budget amounts across different financial categories.
- The Region-wise charts provided an overview of actual expenditure across different regions.
- The Payment Method chart showed the distribution of actual expenditure across Card, Cash, UPI, and Bank Transfer.
- The Year-wise Budget vs Actual chart helped compare financial performance across different years.
- Interactive slicers allowed users to filter dashboard results by year, department, region, category, and payment method.
- The dashboard showed approximately 795M Total Budget and 890M Total Actual Amount based on the overall dashboard data.
- The Budget Utilization was approximately 112%, indicating that actual expenditure was higher than the allocated budget.
- The dashboard presented financial information in a clear and interactive format, supporting faster and more informed financial decision-making.

Conclusion

This experiment demonstrated how Power BI can be used to design an interactive Finance KPI Dashboard for financial reporting and analysis. The dashboard combines multiple visualizations to monitor budget allocation, compare actual expenditure, analyze budget variance, evaluate department and category performance, and understand regional and payment-method spending patterns.

Interactive filters improve the user experience by allowing customized financial analysis. The dashboard provides valuable insights into budget utilization and spending performance, helping management identify overspending areas and support effective financial planning and decision-making.

Learning Outcomes

- ✓ Understand the importance of interactive dashboards in financial business intelligence.
- ✓ Import and prepare the Budget vs Actual Financial Dataset in Power BI.
- ✓ Verify data quality by checking data types, duplicates, missing values, and transaction records.
- ✓ Create KPI Cards to summarize financial performance.
- ✓ Create DAX measures for budget, actual expenditure, variance, utilization, and transactions.
- ✓ Create Clustered Charts to compare budget and actual amounts.
- ✓ Create Donut Charts to analyze payment methods and regional expenditure.
- ✓ Create Bar Charts to analyze department-wise budget variance.
- ✓ Create Treemaps to analyze category-wise budget allocation.
- ✓ Create Year-wise charts to monitor financial trends.
- ✓ Use slicers to build interactive financial reports.
- ✓ Apply professional color contrast, formatting, and visualization techniques to design an official-looking Power BI dashboard.
- ✓ Analyze budget utilization and identify potential overspending.
- ✓ Gain practical experience in developing business intelligence dashboards for financial performance analysis.

Github link: https://github.com/lavanyae4319/Data_Analytics_Assignments